

Question 3: Methods and Concepts

A self-contained, from-the-ground-up technical guide

SIMC 2026 Endeavour Challenge
Design matrix, covariance and Gram matrices · PCA · clustering
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Contents

1. Overview
2. The data and notation
3. Task 3a — Cleaning the data: norms, gaps and masks
4. Task 3b — Variance, mean-centering and the covariance matrix
5. Task 3c — Eigen-decomposition and PCA: the eigenspectrum
6. Task 3d — The leading eigenvector v_1
7. Task 3e — Clustering in covariance space
8. Glossary of terms
9. Appendix — anticipated examiner questions

1. Overview

This guide explains the methods and concepts behind Question 3 of the SIMC 2026 Endeavour Challenge from first principles. It assumes very little prior mathematics: every technical term is defined where it first appears, together with the precise details — formulas, conventions and the reasons behind each modelling choice — that an examiner may ask about. A running example, the analysis of 1000 infrared spectra, keeps the ideas concrete.

The task is a complete data-analysis workflow. We are given a table of spectroscopic measurements; we clean it (3a), quantify how each frequency varies (3b), find the few directions that capture the structure (3c), interpret the most important of those directions (3d), and use them to discover the natural groups of samples (3e). Each step introduces one core idea of applied linear algebra, and the steps build on one another, so the document is best read in order.

2. The data and notation

The design matrix. The data form a table called the **design matrix** X with $N = 1000$ **rows** (each a **sample** — one vial) and $M = 3401$ **columns** (each a **feature** — one infrared frequency). The entry X_{ij} is the absorbance of sample i at frequency j . One row is a **spectrum**; one column is a single frequency recorded across all samples.

Vectors and matrices. A **vector** is an ordered list of numbers (a row or column); a **matrix** is a rectangular grid of numbers. The **transpose** X^T swaps rows and columns. The **dot product** $a \cdot b$ multiplies matching entries and sums them, and is large when two vectors point the same way. These few objects and operations are the entire toolbox for Question 3.

3. Task 3a — Cleaning the data: norms, gaps and masks

Real measurements contain mistakes, so we remove them first. Task 3a finds two kinds of corrupted vials.

Error 1: empty vials. Some vials held no sample, so the instrument recorded only faint background noise — a nearly flat spectrum of small values. To detect these efficiently we reduce each spectrum to a single number using the **Euclidean norm**: for a vector $x = (x_1, \dots, x_M)$, $\|x\| = \sqrt{x_1^2 + x_2^2 + \dots + x_M^2}$. Geometrically the norm is the length of the arrow from the origin to the point x — a one-number measure of overall size. A genuine spectrum has tall absorption peaks and hence a large norm; an empty vial carries only small noise and hence a small norm. Because the norm squares every entry before summing, a few large peaks dominate it, which is exactly why it separates peak-bearing spectra from flat noise so sharply.

A data-driven threshold. We computed all 1000 norms and sorted them. The sorted values reveal a clear **gap**: about 40 vials sit at norm ≈ 1.6 , then nothing until the real vials begin at ≈ 4.3 . Instead of hard-coding a cut-off (which would be arbitrary), we place the threshold in the middle of the largest gap. This is robust — a small change in the data does not move the decision — and leaves no magic constant to justify. The 40 flagged vials turn out to be the last 40 rows of the file, a hint that the corrupted

measurements were appended deliberately.

Error 2: high-norm outliers. The problem hinted at a second error. The loud end of the sorted norms shows another clearly separated group: 36 vials at norm ≈ 8.9 , cut off by a gap about five times wider than any gap inside the bulk, and forming a contiguous block (rows 924–959) immediately before the empties. We flag these as the second error, while being explicit that such a coherent group could instead be a genuinely strong sample; we therefore provide independent evidence (below) and keep the decision reversible with a single switch in the code.

Boolean masks, and why we keep rather than delete. A **boolean mask** is an array of True/False flags, one per vial. We build two — *empty* and *high* — and form the selection with set logic: $\text{keep} = (\text{NOT empty}) \text{ AND } (\text{NOT high})$. We then index the data with this mask rather than deleting rows, for three concrete reasons. (1) **Non-destructive:** the original matrix is never altered, so any choice can be reversed. (2) **Composable:** independent criteria combine in a single expression. (3) **Index-preserving:** vial 193 remains vial 193, so labels are never lost in later steps. Applying the mask leaves 924 clean spectra.

Validation. Three independent checks confirm the removal. (i) Visual: removed spectra are flat noise, kept spectra show sharp peaks. (ii) Statistical: empty vials are $\approx 50\%$ exact zeros, while real spectra are only $\approx 25\%$. (iii) Structural: in the reduced picture of 3c the high-norm group forms one tight isolated cluster, and removing it lowers the cluster count by exactly one ($26 \rightarrow 25$) while leaving every other group intact.

Why exactly 50% zeros? Background noise wobbles randomly around zero, but an instrument cannot record negative absorbance, so every negative wobble is set to zero (it is **clipped**, or rectified). Since about half of a zero-centred wobble is negative, about half of an empty vial's readings become exactly zero — a tell-tale fingerprint that peak-bearing real spectra, whose values are pushed well above zero by genuine absorption, simply do not show.

4. Task 3b — Variance, mean-centering and the covariance matrix

With clean data we ask which of the 3401 frequencies actually distinguish samples. A frequency that reads the same for every vial is uninformative; one that varies is useful. The quantity that measures this is variance.

Mean and mean-centering. The **mean** of feature j is its average over the $\tilde{N} = 924$ kept vials, $\bar{X}_j = (1/\tilde{N}) \sum_i X_{ij}$. **Mean-centering** subtracts this from every entry, giving the centered matrix $\tilde{X}$ with $\tilde{X}_{ij} = X_{ij} - \bar{X}_j$, so each feature now averages zero. We center because we care about how samples *differ*, not about the large average spectrum they share; without centering, the dominant pattern found later would simply be that shared average and would carry no discriminating information. The intuition: in a class where everyone is about 140 cm tall, the informative quantity is who is taller or shorter than average, not the 140 cm everyone has in common.

Variance. The **variance** of feature j is the mean squared deviation from its mean, $\text{Var}_j = (1/\tilde{N}) \sum_i (X_{ij} - \bar{X}_j)^2$. Squaring makes every deviation positive and weights large excursions heavily, so variance is large precisely when a feature swings widely across vials. A precise convention matters here: we divide by $\tilde{N}$ (the *population* variance, NumPy's default $\text{ddof} = 0$), which matches the covariance definition below; dividing by $\tilde{N}-1$ (the *sample* variance, $\text{ddof} = 1$) would not, and would make the trace check below disagree.

The covariance matrix. The **covariance** of two features measures whether they vary together. Assembled for all pairs they form the **covariance matrix** $C = (1/\tilde{N}) X^T X$, an $M \times M$ table with C_{ij} the covariance of features i and j . An off-diagonal entry is positive when features i and j tend to rise and fall together across vials, negative when one rises as the other falls, and near zero when they are unrelated. The diagonal is the special case $i = j$ — a feature's covariance with itself, which is exactly its variance. C is also **symmetric** ($C_{ij} = C_{ji}$), a property we rely on in 3c. To answer 3b we need only the diagonal, which we compute directly rather than forming the whole $M \times M$ matrix.

Trace and the sanity check. The **trace** of a square matrix is the sum of its diagonal entries; here $\text{tr}(C) = \sum_j \text{Var}_j$ is the **total variance** of the dataset. Its invariance is not a coincidence: rotating the feature axes (an orthogonal change of basis) redistributes variance among the new axes but can neither create nor destroy it, so the total is preserved — the same conservation idea as the conserved quantity $p_1 + p_2$ in Question 1. The trace must also equal the sum of the eigenvalues found in 3c; we verified $\text{tr}(C) \approx 7.90$.

Result. Plotting Var_j against j shows the variance is concentrated: most frequencies are flat, while a few narrow **bands** spike sharply. Those bands are the molecular absorption regions where samples differ most, and the single highest-variance frequency is $j = 910$. The implication is that only a handful of frequencies carry information — the data is effectively **low-dimensional**, the structure PCA exploits next.

5. Task 3c — Eigen-decomposition and PCA: the eigenspectrum

Each spectrum has 3401 numbers, yet perhaps only a few independent **patterns** explain almost all the variation. Finding them is **Principal Component Analysis (PCA)**, performed by the eigen-decomposition of the covariance matrix C .

Eigenvectors and eigenvalues. For a square matrix C , a non-zero vector v is an **eigenvector** if $C v = \lambda v$ — multiplying by C only stretches v without rotating it — and the scalar λ is its **eigenvalue**. For a covariance matrix this has a precise meaning: the variance of the data along a unit direction w equals $w^T C w$ (the **Rayleigh quotient**), and for an eigenvector this is exactly λ . Hence each eigenvalue is the variance captured along its eigenvector; the eigenvectors are the principal patterns and the eigenvalues rank them.

Why it is guaranteed to work. Two properties of C make the analysis safe. By the **spectral theorem**, a real symmetric matrix has only real eigenvalues and a full set of

mutually perpendicular (**orthonormal**) eigenvectors — a clean set of axes. And C is **positive semi-definite (PSD)**: $w^T C w \geq 0$ for every w (a variance is never negative), which forces every eigenvalue $\lambda \geq 0$, so each is a genuine variance.

Computing it: `eigh`, not `eig`. NumPy's `eig` handles any square matrix but can return complex values from round-off; **`eigh`** is specialised for symmetric matrices, guaranteeing real eigenvalues and orthonormal eigenvectors while being faster and more stable. We use `eigh`, reverse its ascending output to put the largest eigenvalue first, and **`clip`** tiny negative values ($\approx -10^{-17}$ round-off) up to zero to respect the PSD property.

The eigenvalue spectrum and the elbow. Plotting the sorted eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ (a **scree plot**) shows them fall steeply then flatten onto a near-constant **noise floor**. The large eigenvalues are real **signal**; the flat tail is **noise**. The transition is the **elbow** k^* , located as the largest **multiplicative** drop between consecutive eigenvalues (a ratio, because the signal/noise boundary is a ratio on the log scale; a difference would merely track the huge λ_1). The biggest drop is $\lambda_8/\lambda_9 \approx 4.1$, giving $k^* = 8$. Not 7, because λ_8 still sits about $4\times$ above the floor (signal); not 9, because λ_9 already lies on the floor (noise). This matches the ~ 10 dominant molecules: few components can only build few independent directions.

Dominance and the spectral gap. The ratio $\lambda_1/\lambda_2 \approx 3.5$ measures how strongly the top pattern dominates the next. A large ratio (a wide **spectral gap** $\lambda_1 - \lambda_2$) signals strong low-dimensional structure, and is exactly the condition that makes the Power Method of Question 2 converge quickly.

Cumulative variance — and a caution. The **explained-variance ratio** of component k is $\lambda_k/\Sigma\lambda$; its running total is the **cumulative variance**. The familiar rule of keeping enough components for 95% of the variance is misleading here: the 8 signal components explain only $\approx 55\%$, and reaching 95% needs $k = 667$ — almost all of them noise — because the small noise floor is spread over ~ 900 dimensions (the **rank** is at most $\tilde{N}-1$). The elbow/spectral-gap criterion ($k^* = 8$) is the honest measure. Finally $\Sigma\lambda \approx \text{tr}(C) \approx 7.90$ confirms the decomposition.

6. Task 3d — The leading eigenvector v_1

PCA found ~ 8 real patterns; Task 3d examines the most important, the **leading eigenvector** v_1 (the eigenvector of the largest eigenvalue λ_1). It is an $M = 3401$ -dimensional vector in **feature space** — the space of the columns of X — assigning a weight (**loading**) to each frequency. Equivalently, v_1 is the single line through the cloud of mean-centred spectra along which their spread is greatest: the **direction of greatest variance**.

Loadings versus scores. Two related vectors should be kept apart. The **loadings** are the 3401 entries of v_1 themselves — how much each frequency contributes to the pattern. The **scores** are the one-number summaries $z_i = \tilde{x}_i \cdot v_1$ obtained by **projecting** each vial onto v_1 . Loadings describe the pattern; scores place each sample along it. Because v_1 is a unit vector, the dot product $\tilde{x} \cdot v_1$ is exactly the signed distance of the sample along the v_1 direction, and among all directions v_1 is the one whose scores

have the greatest variance — the most information-preserving single summary of a vial.

A contrast direction. Plotting the loadings shows weight concentrated in two narrow bands ($\approx$ frequencies 900–960 and 3150–3260) and near zero elsewhere — the informative bands of 3b. The loadings take both signs ($\sim 27\%$ negative). Mixed signs make v_1 a **contrast**: samples scoring high in the positive bands score low in the negative ones. So v_1 measures the *balance* of absorption between two frequency groups, not overall brightness, and the sign of a vial's score tells you which side of that balance it falls on.

The sign ambiguity. If $C v = \lambda v$ then $C(-v) = \lambda(-v)$, so an eigenvector is defined only up to an overall $\pm$ sign (a line carries no built-in arrow). The solver may return either; we flip v_1 so its largest loading is positive. This changes nothing mathematically and only makes the plot easier to read.

Cosine similarity. To test whether v_1 resembles a real measurement we compare shapes with **cosine similarity**, $\cos(a,b) = (a \cdot b) / (\|a\| \|b\|)$ — the cosine of the angle between two vectors (+1 identical, 0 perpendicular, -1 opposite). It is **scale-invariant**, comparing direction not magnitude, which is appropriate because v_1 is unit length while spectra are not; Euclidean distance would instead be dominated by size differences and answer the wrong question.

Result, explained by mean-centering. v_1 matches the *centered* spectrum of vial 193 with cosine 0.92, but only 0.67 against that vial's *raw* spectrum and ≈ 0.03 against the average spectrum. The reason: v_1 lives in the centered space and is essentially **orthogonal** (perpendicular) to the average spectrum. Raw spectra are dominated by that shared average, so v_1 cannot resemble them; it instead resembles a **difference-spectrum** — how the most-deviated sample differs from the mean. This is precisely what mean-centering was designed to expose, and it mirrors Question 1, where the dominant eigenvector was the system's long-term steady state.

Why this matters downstream. Because v_1 is the single most discriminating axis, its score $z = \tilde{x} \cdot v_1$ is the best one-number label for a vial. In Task 3e we extend the idea to the four leading eigenvectors at once, giving each vial four coordinates that, as we shall see, separate the recipes cleanly — so the interpretation built here is the foundation for the clustering that follows.

7. Task 3e — Clustering in covariance space

Having identified the few directions that carry the structure, Task 3e uses them to reveal the natural groups among the vials. This is **unsupervised** learning: we are given no labels saying which vial holds which drink, so we must let the geometry of the data alone expose the groups.

Dimensionality reduction by projection. We keep the four leading eigenvectors as the columns of an $M \times 4$ matrix $V_4 = [v_1 \ v_2 \ v_3 \ v_4]$ and project the data onto them: $Z = \tilde{X} V_4$, of size $\tilde{N} \times 4$. Each row $z_i = (\tilde{x}_i \cdot v_1, \dots, \tilde{x}_i \cdot v_4)$ holds that vial's four **scores** — its coordinates in the top-4 **principal subspace**. This is **dimensionality reduction**: every spectrum's 3401 numbers are compressed to just 4 that retain the most

important variation. We choose four because they lie within the 8 signal directions of 3c yet are few enough to inspect by eye, and because the problem asks for the top four.

The scatter-plot matrix. Four coordinates give $C(4,2) = 6$ distinct pairs (PC1-PC2, PC1-PC3, ..., PC3-PC4), which we display as six 2-D scatter plots. Each **point** is one vial placed at its two scores. We cannot view four dimensions at once, so we inspect every 2-D shadow of the 4-D cloud: a group that happens to overlap in one projection often separates cleanly in another, so reading all six panels together gives the most reliable count. We draw points small and semi-transparent (size 10, alpha 0.4) so that where many vials overlap the colour deepens, keeping dense regions visible rather than saturating into a solid blob.

Clusters. A **cluster** is a group of points lying close together and well separated from other groups — here, vials whose spectra are nearly identical, i.e. the same **concoction**. To count them without external libraries we use simple **leader clustering**: walk through the points and start a new cluster centre whenever a point is farther than a radius r from every existing centre. The only choice is r ; setting it to a few times the typical nearest-neighbour spacing gives a **stable** count of 25 across a wide range (r from $6\times$ to $12\times$), so the answer is a property of the data, not of the threshold.

Result and meaning. The 924 cleaned vials fall into about **25 distinct clusters** — roughly 25 unique recipes — comfortably below the ‘fewer than 100 unique samples’ stated in 3h. It is worth appreciating what just happened: the eigenvectors $v_1\dots v_4$ live in *feature* (frequency) space, yet projecting the samples onto them reveals structure in *sample* space, because vials with similar spectra receive similar scores. Four numbers out of 3401 are enough to separate the recipes cleanly — a direct payoff of the low-dimensionality found in 3c, and a concrete demonstration that the eight signal directions really do encode the chemistry that distinguishes the drinks. Task 3f will repeat this using the Gram matrix, whose sample-space eigenvectors share the same non-zero eigenvalues (Question 2d), and should recover the same groups. In short, Task 3e turns the abstract eigen-directions of 3c into a concrete, countable map of what the dataset actually contains.

8. Glossary of terms

Sample / feature. A sample is one measured object (a vial / row of X); a feature is one measured quantity (a frequency / column of X).

Design matrix X . The full data table: $N = 1000$ samples (rows) by $M = 3401$ features (columns).

Vector / norm. A vector is an ordered list of numbers; its Euclidean norm $\|x\| = \sqrt{\sum x_i^2}$ is its length, a one-number measure of overall size.

Transpose (X^T). The matrix with rows and columns swapped: $(X^T)_{ij} = X_{ji}$.

Dot product. $a \cdot b = \sum a_i b_i$; multiply matching entries and add. Measures alignment of two vectors.

Boolean mask. An array of True/False flags used to select rows, combined with NOT ($\sim$) and AND ($\&$).

Mean-centering. Subtracting each feature's average so every column averages zero; focuses analysis on differences, not the shared average.

Variance. Mean squared deviation from the mean; divide by N ($\text{ddof}=0$) to match the covariance definition.

Covariance matrix C . $C = (1/\tilde{N}) X^T X$ ($M \times M$, symmetric); off-diagonals show which features co-vary, the diagonal holds the per-feature variances.

Trace. Sum of a matrix's diagonal entries; for C it is the total variance and equals the sum of the eigenvalues.

Eigenvector / eigenvalue. A direction v with $C v = \lambda v$ (C only stretches it); λ is the variance captured along v .

Spectral theorem. Every real symmetric matrix has real eigenvalues and an orthonormal eigenvector basis.

Positive semi-definite (PSD). $w^T C w \geq 0$ for all w ; guarantees all eigenvalues ≥ 0 .

eigh vs eig. `eigh` is the symmetric-matrix solver (real eigenvalues, orthonormal eigenvectors, faster, stabler); `eig` is the general-purpose routine.

PCA. Principal Component Analysis: the eigen-decomposition of the covariance matrix; finds the directions of greatest variance.

Scree plot / elbow. Plot of sorted eigenvalues; the elbow k^* separates large 'signal' eigenvalues from the flat 'noise' floor.

Spectral gap. $\lambda_1 - \lambda_2$ (or the ratio λ_1/λ_2); a wide gap means one direction dominates and the Power Method converges fast.

Explained / cumulative variance. $\lambda_k / \sum \lambda$ is the fraction explained by component k ; its running total is the cumulative variance.

Leading eigenvector v_1 . The eigenvector of the largest eigenvalue; the single direction of greatest variation.

Projection / score. $z_i = \tilde{x}_i \cdot v$, the dot product of a spectrum with an eigenvector; a coordinate in the principal subspace.

Dimensionality reduction. Replacing each sample's many features by a few principal-component scores that retain the most variation.

Cluster. A group of points close together and separated from others; here, vials of the same concoction.

Cosine similarity. $\cos(a,b) = (a \cdot b) / (\|a\| \|b\|)$; the cosine of the angle between vectors (+1 same, 0 perpendicular); scale-invariant.

Orthogonal. Perpendicular; two vectors are orthogonal when their dot product is 0.

9. Appendix — anticipated examiner questions

1. Why transpose the data file first?

It loads as 3401×1000, but the problem defines samples as rows (1000 vials). Transposing puts one vial per row so every later axis operation is correct.

2. The problem says 3041 features but the data has 3401 — which is right?

The file genuinely has 3401 columns; 3041 is a digit-swap typo. We trust the data, use 3401, and key off the length-1000 axis to find the samples.

3. What is the row norm and why does it detect empty vials?

$\|x\| = \sqrt{\sum x_j^2}$ compresses a 3401-number spectrum to one 'overall size'. Empty vials (noise only) are small; real spectra have peaks and are large.

4. How did you choose the empty-vial threshold? Why not a fixed value?

We placed it in the middle of the largest gap in the sorted norms — data-driven, with no arbitrary constant; the empties fall naturally below the gap.

5. How do you know the 40 are empty, not weak real samples?

A wide clean gap (≈ 1.7 then a jump to ≈ 4.3), flat noisy spectra, and $\approx 50\%$ exact zeros (clipped zero-mean noise) versus $\approx 25\%$ for real spectra.

6. Why treat the 36 high-norm vials as an error — could they be a real strong sample?

They might be; we flagged them for the $\sim 5\times$ wider gap and contiguous-block placement. We showed they form one isolated cluster, noted the alternative, and kept the choice reversible.

7. How did you validate the removals?

Visually (flat vs peaks), statistically ($\approx 50\%$ zeros), and structurally (removing the high group drops exactly one isolated cluster, 26→25, leaving the rest intact).

8. Why use a keep-mask instead of deleting rows?

It is non-destructive (original intact), composable (combine $\sim$ empty & $\sim$ high in one step), and preserves original row indices for traceability.

9. What does mean-centering do and why is it necessary?

It subtracts each feature's mean so analysis is about differences, not the shared average spectrum; otherwise the top PCA direction would just be that average.

10. Why does the diagonal of C equal the variance?

After centering, $C_{jj} = (1/\tilde{N})\sum_k X_{kj}^2$, which is exactly the definition of feature j 's variance.

11. Why np.var with ddof=0 (divide by N), not N-1?

The covariance is defined with a $1/\tilde{N}$ factor; ddof=0 (population variance) matches it, whereas ddof=1 (sample variance) would not.

12. What is the trace of C and why check it?

Sum of the diagonal = total variance; it must equal the sum of eigenvalues, and matching them (≈ 7.90) confirms the decomposition.

13. Why `eigh` instead of `eig`?

C is symmetric; `eigh` guarantees real eigenvalues and an orthonormal eigenvector basis and is faster/stabler. `eig` is general-purpose and may return complex round-off.

14. In plain terms, what is an eigenvalue here?

The variance of the data along its eigenvector direction ($\mathbf{v}^T \mathbf{C} \mathbf{v} = \lambda$). Big = captures much of the variation; tiny = noise.

15. Why clip negative eigenvalues to zero?

C is PSD, so true eigenvalues are ≥ 0 ; tiny negatives are floating-point round-off, and clipping keeps them physically valid.

16. How did you pick the elbow $k^* = 8$, and why not 7 or 9?

Largest multiplicative drop between neighbours, $\lambda_8/\lambda_9 \approx 4.1$. λ_8 is $\sim 4\times$ above the noise floor (signal); λ_9 already sits on the floor (noise).

17. What does $\lambda_1/\lambda_2 \approx 3.5$ tell you?

One direction dominates — strong low-dimensional structure and a wide spectral gap, which also makes the Power Method converge fast.

18. Why is the 'keep 95% of variance' rule misleading here?

The 8 signal components explain only $\approx 55\%$; reaching 95% needs ≈ 667 mostly-noise components because the broad noise floor inflates the total. The elbow is the honest measure.

19. Why eigendecompose the $M \times M$ covariance, not the smaller $N \times N$ Gram matrix?

They share nonzero eigenvalues, but we need the feature-space eigenvectors $\mathbf{v}_k$ for 3d/3e, which `eigh(C)` returns directly; the ~ 14 s cost is paid once and reused.

20. What does the leading eigenvector $\mathbf{v}_1$ represent?

The single mix of frequencies along which samples differ most; the projection $\tilde{\mathbf{x}} \cdot \mathbf{v}_1$ is the most informative one-number summary of a vial.

21. Why does $\mathbf{v}_1$ have negative components?

It is a contrast direction: positive vs negative bands. It separates samples by the balance of absorption between frequency groups, not overall brightness.

22. Why cosine similarity, not Euclidean distance, to compare $\mathbf{v}_1$ with spectra?

We care about shape, not size; $\mathbf{v}_1$ is unit length and spectra are not. Cosine is scale-invariant and measures the angle between directions.

23. Why does $\mathbf{v}_1$ match a centered spectrum (0.92) but not a raw one (0.67) or the mean (0.03)?

v_1 lives in the centered space and is $\approx$ orthogonal to the mean spectrum. Raw spectra are dominated by that mean, so v_1 resembles difference-spectra, not raw ones.

24. In 3e, why project onto exactly four principal components?

Four lie within the eight signal directions of 3c yet are few enough to view as the six pairwise scatter plots; they retain the dominant structure while discarding noise.

25. In the 3e scatter plots, what does a point represent, and why small transparent markers?

Each point is one vial at its two PC scores. Small, semi-transparent markers ($s=10$, $\alpha=0.4$) stop dense clusters from saturating into a solid blob, so overlap stays visible.

26. How did you count clusters without scikit-learn, and is it robust?

Leader clustering: walk the points and start a new centre whenever one is farther than radius r from all centres. The count is 25 and stable for r from $6\times$ to $12\times$ the typical neighbour spacing.

27. How can feature-space eigenvectors reveal groups among the samples?

By projection: each sample's scores $\tilde{x} \cdot v_k$ are its coordinates in the principal subspace. Samples with similar spectra get similar scores, so sample-space clusters appear even though the axes are feature-space directions.

28. How does Task 3 connect to Questions 1 and 2?

Q1: the dominant eigenvector is the system's long-term state; here it is the main axis of variation. Q2: the Power Method finds it iteratively, and a wide spectral gap makes it fast.

29. What is the difference between the covariance and Gram matrices?

Covariance $\tilde{X}^T \tilde{X} / \tilde{N}$ ($M \times M$) lives in feature space and says which features co-vary; Gram $\tilde{X} \tilde{X}^T / \tilde{N}$ ($N \times N$) lives in sample space and says which samples are similar. They share nonzero eigenvalues.

30. Could the cleaning have removed a real cluster by accident?

For empties, no — they are flat noise. For the high-norm group, possibly; that is why we proved it is one isolated cluster, showed removal does not disturb the others, and made it reversible.